OverFlow: A Geospatial Decision Support System for Flood Evacuation Planning

CAPSTONE PROJECT

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ABSTRACT

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1 Introduction

Flood-evacuation planning brings together decisions about affected communities, usable roads, shelter space, and transport. A road closure can change which center is reachable; a full center can change the route or leave some evacuees unassigned. Planners need to examine these conditions together.

This study develops *OverFlow: A Geospatial Decision Support System for Flood Evacuation Planning* for Laoag City, Ilocos Norte. The system uses an interactive three-dimensional (3D) environment to help local disaster personnel configure and compare evacuation scenarios.

1.1 Project Context

Laoag City Hall has provided terrain, barangay, road, population, and flood-susceptibility records for the study. These layers locate communities and infrastructure, but a planning decision also depends on event-specific demand, road access, center capacity, and available transport. OverFlow connects the geographic records with those operational conditions.

The system treats population figures as references for setting evacuation demand and the supplied flood-susceptibility map as background hazard information. SFINCS provides the rainfall-driven flood-scenario method. Keeping these sources distinct allows planners to understand what a scenario represents and compare its resulting evacuation arrangements.

1.2 Purpose and Description

OverFlow enables planners to select any of Laoag City's 80 coded barangays, individually or in groups. Hovering highlights a barangay boundary; clicking selects or deselects it. The

system presents geographic layers in 3D and overhead views.

For each scenario, the planner specifies expected evacuees, rainfall conditions, road and bridge closures, center availability and capacity, vehicle resources, and travel assumptions. The analysis identifies feasible routes and center assignments, estimates travel and vehicle requirements, and reports demand that cannot be served. Scenarios can be saved, reopened, compared, and exported for planning discussions. Modeled road exposure is presented for review; a road is excluded from routing only when the planner marks it closed.

1.3 Objectives of the Study

The general objective is to develop and evaluate OverFlow as a geospatial tool for configuring and comparing flood-evacuation scenarios in Laoag City. Specifically, the study aims to:

1. integrate relevant terrain, barangay, road, population, center, and resource data;
2. provide barangay-based scenario controls for demand, flooding, road access, shelter capacity, and transport;
3. develop SFINCS flood scenarios and evaluate their local applicability;
4. calculate feasible routes and capacity-constrained center assignments, including unserved demand and resource estimates;
5. present and compare scenario inputs and results in an interactive geographic interface; and
6. evaluate system functions and their usefulness in planning tasks.

1.4 Significance of the Study

For disaster planners and responders, OverFlow provides a common view of how changing road access, demand, shelter capacity, and transport affect an evacuation arrangement. Reporting unserved demand makes capacity and connectivity constraints visible. Saved scenarios support review and discussion across offices.

Residents are indirect beneficiaries of better-informed preparedness planning. The study also provides researchers and developers with an evaluated example of combining local geographic data, flood scenarios, and constrained evacuation analysis in one planning tool.

1.5 Scope and Delimitations of the Study

The interface covers all 80 coded Laoag barangays. Barangays 38-A Mangato East, 38-B Mangato West, and 39 Sta. Rosa form the technical SFINCS pilot area. The geographic extent of the final flood-model evaluation will follow the available terrain, boundary data, and observations from a dated flood event. Interface coverage does not imply that flood results have been validated citywide.

The study covers scenario configuration, flood modeling, geographic display, road and center analysis, travel and vehicle estimates, and scenario comparison. Population figures are reference values; the planner supplies expected evacuees and usable capacities for each scenario. Flood exposure informs review, while planner-confirmed closures determine which roads the route analysis excludes.

The supplied elevation grid has approximately 91 m cell spacing, so detailed street-level flood claims require finer terrain and local evaluation. Outputs are planning estimates under

stated assumptions. The system is designed for preparedness analysis rather than real-time forecasting, public navigation, or emergency dispatch.

1.6 Operational Definition of Terms

Evacuation Demand. The expected number of evacuees specified for a selected barangay in a scenario.

Evacuation-Center Allocation. Assignment of evacuees to reachable centers within the usable capacities set for a scenario.

Feasible Evacuation Route. A road-network path from an origin to a center under the access conditions specified in a scenario.

Flood-Evacuation Scenario. A set of selected barangays, rainfall and demand assumptions, road access, center availability and capacity, and transport resources used to examine an evacuation arrangement.

Unserved Demand. Evacuees who cannot be assigned to a reachable center within the available capacity.

Usable Capacity. The number of evacuees a center can accommodate under the conditions specified for a scenario.

2 Review of Related Literature

2.1 Introduction

Flood-evacuation planning connects hazard assessment, road access, shelter assignment, and human movement. This review examines how these concerns have been represented in geospatial systems and identifies the methods most relevant to a city-level planning tool.

2.2 Flood-Aware Road Accessibility and Geospatial Decision Support

Li et al.'s review of urban flood evacuation research places hazard assessment before shelter selection, route development, and movement analysis [1]. Alabbad et al. showed in two Iowa cities that inundation can disconnect roads and bridges, lengthen journeys, and change the nearest reachable emergency facility [2]. Together, these studies establish why route analysis must use the access conditions of a flood scenario rather than the ordinary road network alone.

GIS studies in Thailand and Nepal combine flood-susceptibility mapping with route and destination analysis [3, 4]. Parajuli et al. also accounted for the effects of slope and flooding on walking speed and compared their susceptibility map with satellite observations. In the Philippines, Bolanio et al. assessed the suitability of existing evacuation centers in Las Nieves and presented the results through a web map [5]. Abuzo et al. compared road conditions under normal and flood scenarios in Cagayan de Oro [6]. These cases show the value of examining destinations and their connecting routes together, using local geographic evidence.

Dynamic-route studies address changes that a static map cannot represent. Ogawa et

al. weighted roads according to present and projected inundation in a Japanese evacuation simulation [7]; Jeon et al. recalculated pedestrian routes when a flooded path or destination became unavailable [8]. Their predictive and real-time functions require hazard data and validation beyond a static road-closure scenario. OverFlow uses modeled exposure to inform planner review and excludes a road from routing when the planner confirms its closure.

2.3 Shelter Capacity and Population Allocation

Shelter choice involves more than proximity. Qiu et al. combined flood-risk assessment, shelter selection, and evacuation simulation, reporting improved outcomes for their modeled shelter arrangement [9]. Sharbaf et al. treated population exposure, shelter risk, and route risk as parts of one shelter-network problem [10]. Wang et al. compared routes and shelters under different road-risk and pedestrian-demand priorities [11]. These studies show that a nearby facility may be unsuitable because of exposure, access, or limited space.

A Philippine location-routing model by de Veluz et al. jointly considered facilities, routes, capacities, travel time, and vehicles across typhoon scenarios [12]. Gama et al. examined how shelter openings, evacuation orders, and assignments change over time as flood and road conditions develop [13]. Both studies demonstrate that assignments depend on the conditions and priorities of a particular scenario.

Other work examines who can reach a shelter and whether enough space is available. Bera et al. found that existing shelters in an Indian case could not serve all households within the tested travel-time limits [14]. Rendana et al. showed that areas classified as low flood risk are not necessarily suitable shelter sites once road access and other conditions are considered [15]. Yin et al. examined the transfer needs of older residents in coastal cities, while Janatabadi and Ermagun identified communities with high flood risk and low

shelter access [16, 17]. Chen et al. explicitly recorded demand that existing shelters could not accommodate [18].

For OverFlow, the initial assignment rule is deliberately explicit: evacuees are assigned only to reachable centers with available capacity, and remaining demand is reported. The literature identifies additional concerns, including shelter risk and mobility needs, that require local data and defined rules before inclusion in the allocation method.

2.4 Evacuation Simulation and Related Systems

Aketa et al. provided an early example of linking flood and road-network analysis in a GIS evacuation system [19]. Mizumura et al. later compared routes that leave inundated areas quickly with ordinary shortest paths, showing that shorter distance and lower flood exposure can lead to different choices [20]. The U.S. Army Corps of Engineers' LifeSim represents a more detailed approach, combining warnings, population movement, traffic, flood exposure, and potential life loss [21]. These systems illustrate a progression from route calculation to dynamic evacuation modeling, with increasing demands for behavioral and hazard evidence.

González-Villa et al. combined GIS, network analysis, and pedestrian and vehicle simulation in a wildfire evacuation system [22]. Du et al. coupled flood, agent-based, and traffic models to compare shelter arrangements and routing strategies; their results showed that shelter distribution and system-level routing can substantially affect modeled clearance time [23]. These systems demonstrate the value of comparing complete arrangements, while their traffic and behavior models require data that a simpler scenario tool does not assume.

Studies of data-constrained and coastal settings reinforce the importance of stating

what a simulation represents. Butters and Dawson tested shelter locations, crossings, and population samples in Kibera and found that a very small sample produced unstable risk estimates [24]. Yang et al. examined warning time, infrastructure failure, and shelter capacity in Shanghai [25]. Bustami et al. showed how compound hazards and local shelter placement changed modeled outcomes in Connecticut [26]. Their numerical results belong to those modeled settings; the transferable lesson is to compare alternatives under documented inputs and examine local as well as citywide effects.

Evacuation also places demands on vehicles and post-arrival supplies. Santana-Robles et al. linked shelter assignment with aid delivery after a Mexican flood [27]. Woo and Kang combined bus evacuation, congestion, shelter selection, and relief routing [28]. Insani et al. modeled vehicle reuse, uncertain demand, and relief delivery before inundation reached affected areas [29]. These studies motivate recording resource assumptions alongside routes and center assignments, without treating a simplified vehicle estimate as a traffic or logistics simulation.

2.5 Terrain Processing and Evacuation Behavior

O'Callaghan and Mark's D8 method derives drainage directions and contributing area from gridded elevation data [30]. It can provide a terrain reference, but flood depth and timing require a flood model. OverFlow uses SFINCS for rainfall-driven scenarios, with its inputs, forcing, and outputs documented by Deltares [31, 32, 33].

Observed behavior also differs from a mathematically shortest path. Lim et al. found that route choice among flood-affected households in Quezon City was influenced by hazard, socioeconomic, and evacuation factors [34]. Borowska-Stefańska et al. showed that self-evacuation can redistribute traffic well beyond a flood zone [35]. OverFlow's routes are

planning options under stated access and travel assumptions, rather than predictions of individual movement or congestion.

2.6 Synthesis and Research Gap

The literature establishes strong methods for flood mapping, road analysis, shelter assessment, and evacuation simulation, but these functions are often studied separately or within data-intensive models. Philippine studies provide evidence on route behavior, center suitability, flooded roads, and location-routing decisions [34, 5, 6, 12]. The reviewed local applications do not combine these concerns in one city-level interface for comparing flood and operational scenarios.

OverFlow addresses this gap by bringing flood-scenario information, confirmed road closures, evacuation demand, reachable-center capacity, and resource assumptions into one planning workflow. Its evaluation examines the flood model, route and assignment constraints, system functions, and staff use as distinct questions. This structure allows the study to report what each component demonstrates without requiring readers to treat a route calculation as proof of road safety or an estimated travel time as a prediction of actual evacuation behavior.

3 Methodology

This chapter describes the design, data sources, instruments, procedures, and analysis used to develop and evaluate OverFlow in Laoag City.

3.1 Research Design

The study uses a descriptive-developmental design. Interviews document how disaster-management personnel identify affected areas, assess access and shelter capacity, and compare evacuation arrangements. An iterative development process translates these requirements into system functions and tests. The work follows the design science stages of problem identification, solution design, demonstration, and evaluation [36].

SFINCS is used for rainfall-driven flood scenarios [31, 32, 33]. Model setup records the terrain, rainfall, boundary conditions, parameters, and spatial and temporal resolution. A dated simulation is compared with independent flooded and dry observations before local flood-accuracy claims are made. Where this comparison is not possible, the output is presented as a technical demonstration.

Evacuation analysis excludes road links marked closed by the planner, calculates paths to reachable centers, and assigns expected evacuees within usable center capacities. NetworkX weighted paths and OR-Tools minimum-cost flow are the implementation directions. Route weights, assignment costs, and travel and vehicle assumptions are specified before testing. Results report unserved demand where access or capacity is insufficient.

Evaluation uses functional and scenario tests, followed by staff planning tasks if the partner office participates. Tests check the configured road and capacity constraints, scenario storage, and displayed results. The study does not compare OverFlow with another software

system.

3.2 Locale of the Study, Population, and Sample

The study area is Laoag City, Ilocos Norte. The interface allows selection among all 80 coded barangays through hover highlighting and click selection. Barangays 38-A Mangato East, 38-B Mangato West, and 39 Sta. Rosa form the technical SFINCS pilot. The area used for observed-event evaluation is determined by the available flood evidence.

Laoag CDRRMO is the proposed partner for interviews and staff evaluation. Personnel involved in evacuation planning or operations will be selected purposively. Approximately six staff members form the provisional interview target; the final participant count and roles will be reported after recruitment.

3.3 Data Gathering Tools

Formal request letters produced an initial City Hall package, largely through the City Engineering Department, containing elevation, barangay boundaries, population, roads, and flood-susceptibility information. Additional requests seek geocoded center and capacity records, current vehicle resources, bridge locations, and observations from dated flood events. LiPAD is a prospective source of finer bare-earth terrain data.

QGIS and Python tools will be used to inspect, align, and prepare geographic layers. Roads will be cleaned and connected for network analysis, and barangay codes will link boundaries to population records. A dataset inventory will record source, date, coverage, coordinate system, completeness, and permitted use. Population counts serve as references; expected evacuees are entered separately for each scenario.

3.4 Research Instruments

A semi-structured interview guide covers planning steps, road closures, center choice, capacity, transport, and comparison of alternatives. A dataset checklist records source quality and preparation decisions. A system-test checklist records each scenario's inputs, expected behavior, observed output, and result.

The staff evaluation form combines task completion, a five-point rating questionnaire, and open comments. Tasks include selecting barangays, changing demand, closing a road, adjusting center capacity, and comparing scenarios. Ratings address the clarity of outputs, ease of editing inputs, and usefulness of comparisons. Instruments and task criteria will be reviewed before use.

3.5 Data Gathering Procedures

The team will coordinate with the proposed partner office, complete the applicable ethics review, and obtain informed consent before interviews or staff evaluation. Interviews will identify planning practices and requirements; audio will be recorded only with separate permission.

The City Hall files and any additional records will be inventoried, checked, aligned, and prepared for system use. Development then proceeds through geographic display, scenario configuration, SFINCS integration, evacuation analysis, and scenario comparison. Each function will be tested against defined inputs and expected behavior.

SFINCS evaluation will examine input alignment, numerical behavior, boundary conditions, and agreement with observations from a dated event. Without suitable terrain, boundary evidence, and observations, its maps will remain labeled technical demonstra-

tions and will not be used to judge road safety. Planners will review modeled exposure and specify road closures for evacuation scenarios.

Functional tests will check barangay selection, closed-road avoidance, center reachability and capacity, unserved demand, and preservation of saved scenarios. Participating staff will then complete planning tasks and provide ratings and comments. The report will present dataset checks, test results, task outcomes, and staff feedback without identifying participants.

3.6 Statistical Treatment of Data

Interview notes and open comments will be grouped into themes and checked against the original responses. Task completion and test outcomes will be reported as counts and percentages, with the number attempted stated. Five-point ratings will be summarized by response frequency and item mean; missing responses will be excluded from that item's mean. These descriptive results will be interpreted alongside staff comments and scenario-test evidence. No inferential test or claim of superiority over another system is planned.

4 Results and Discussion

4.1 System Development and Scenario-Analysis Results

4.2 System Evaluation Results

5 Conclusions and Recommendations

5.1 Conclusion

5.2 Recommendation

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APPENDICES

A Detailed Tables

B Questionnaires